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Global Facilities - Design & Construction Standard -I&C - Naming Convention

ECN Facility: GLOBAL FACILITIES

ECN Area: ENGINEERING

Owner(s) ENGINEERING

Target Audience: All Facilities

Document Type: Standard

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[ECN History](#)[Submit a Change Request](#)

1.0 Purpose

This Standard defines a common procedure for identification of instrument, equipment, and system as minimum requirement in in Micron greenfield project design.

Any deviation from this standard shall be submitted for approval during evaluation. Deviation list format shall refer to Appendix A in this standard document.

2.0 Scope

2.1 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites for new Greenfield FAB Construction projects.

2.2 This document applies to design document and drawings, such as P&ID equipment, I/O tags, Instrument tags, SCADA, etc.

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3.0 Definitions and Acronyms

- **EHS** - Environmental, Health, and Safety
- **I&C** - Instrumentation & Control
- **FMCS** - Facility Management and Control System
- **P&ID** - Piping and Instrument Diagram
- **PLC** - Programmable Logic Controller
- **IO** - Input Output
- **RIO** - Remote IO
- **SAT** - Site Acceptance Test
- **SCADA** - Supervisory Control and Data Acquisition
- **SOO** - Sequence of Operation
- **ANSI** - American National Standards Institute
- **NFPA** - National Fire Protection Association
- **IEC** - International Electrotechnical Commission
- **ISA** - International Society of Automation
- **LPOC** - Lateral Point of Connection

4.0 Roles and Responsibilities

Roles	Responsibilities
GFTT Lead (Discipline Engineer)	• Develop and maintain System Standard document • Ensure Best Known Methods (BKM), lessons learned, and emerging technologies are documented in System Standard • Perform periodic document review and update System Standard
FCT Lead (Discipline Engineer)	• Review and provide technical feedback on System Standard • Support the Site teams by providing knowledge and technical expertise
Site Project Construction Teams	• Review and provide technical feedback on System Standard • Incorporate System Standard into Project Engineering, Design, and Construction • Document project deviations, exceptions, and exclusions from System Standard • Share new lessons learned to GFTT Lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard
Site Facilities Teams	• Use System Standard as technical reference

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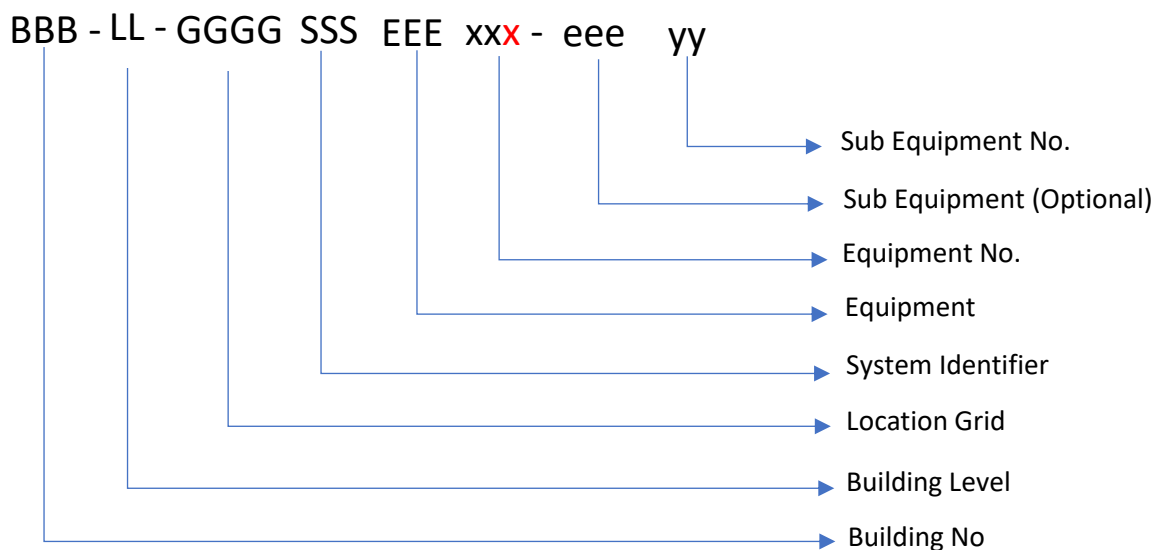
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5.0 Applicable Codes and Standards

- a. ASME (ANSI) A13.1 Scheme for the Identification of Piping Systems.
- b. International Fire Code
- c. International Mechanical Code
- d. NFPA 704 Standard System for the Identification of the Hazards of Materials for Emergency Response.
- e. IEC 60051 Direct acting indicating, measuring instruments
- f. IEC 60417 Graphical symbols for use on equipment.
- g. IEC 60445 Identification of equipment terminals.
- h. ISA 5.1 (2009) Instrumentation symbols and identification
- i. ISA 5.5 Graphic symbols for process displays.

6.0 Identification

- 6.1 Process and Mechanical equipment shall be assigned and labelled as such:



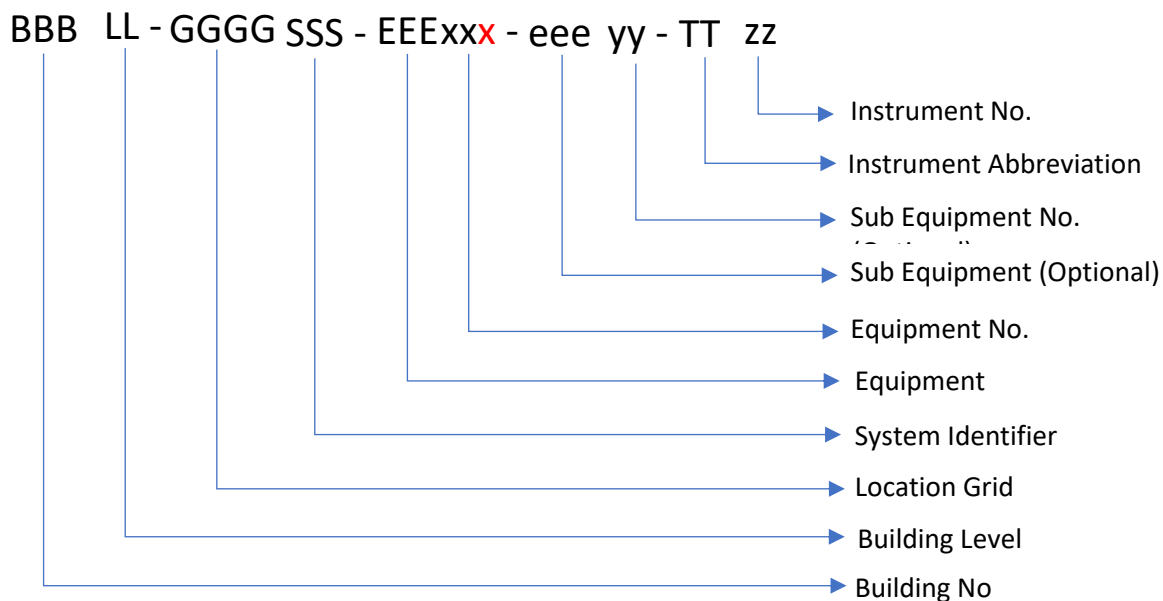
Location grid shall refer to North-West grid of the object/equipment.

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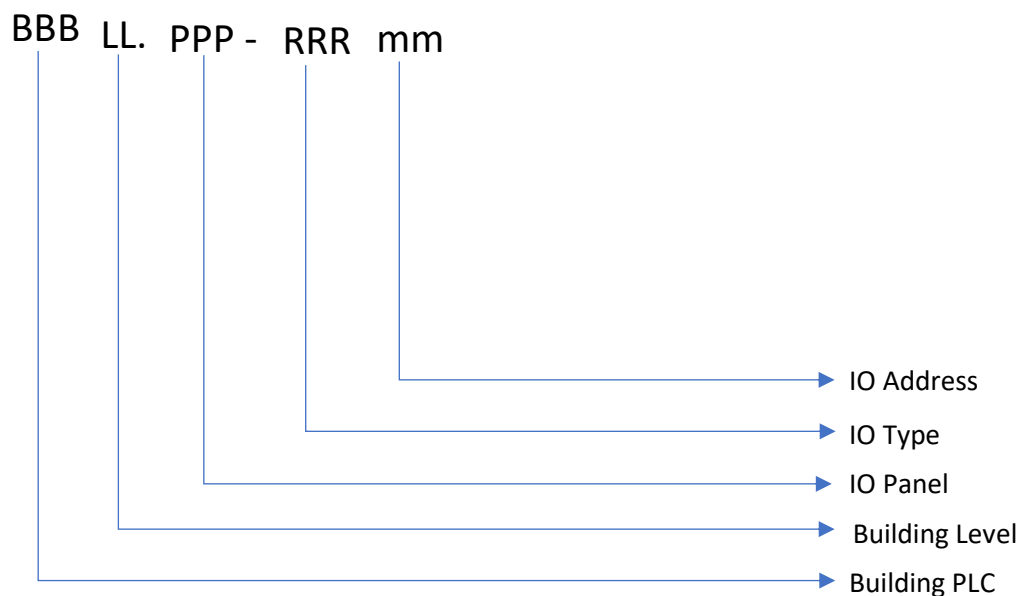
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6.2 Instrument numbers shall be assigned in accordance as such :



6.3 Additional Tag to Instrument label at signal cables can refer to the following recommended label:

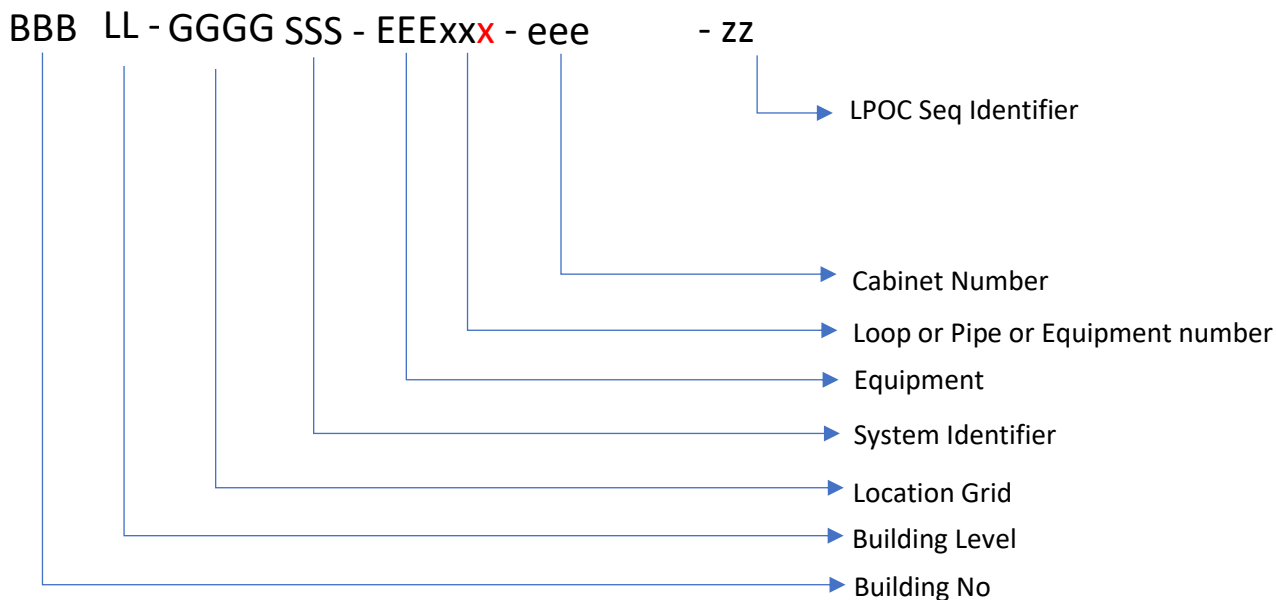


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6.4 Lateral Point of Connection Tagging shall be assigned in accordance as such :



Examples

Building Identifier	Building Level	Grid Location	System Identifier	Equipment Type	Loop or Pipe No.	Cabinet No.	LPOC Seq. Identifier							
1	2	3	4	5	6	7	8							
Building Identifier	-	Building level	-	Grid Line Location	-	System Identifier	-	Equipment Type	-	Loop or Pipe Number	-	Cabinet Number	-	Lateral Point of Connection (LPOC) Sequence Number
C10A2	-	10	-	F29N	-	H2O2-30%	-	SCME	-	-	-	01	-	-
C10A2	-	10	-	F29N	-	IPA	-	CCB	-	-	-	01	-	-
F10A2	-	20	-	F29T	-	NH4OH-29%	-	VMB	-	2	-	02	-	-
F10A2	-	20	-	F29N	-	IPA	-	VMB	-	1	-	06	-	-
F10A2	-	20	-	F43T	-	IPA	-	VMB	-	3	-	01	-	-
C10A2	-	10	-	F29N	-	PCDDS	-	EMOPNL	-	-	-	01	-	-
C10A2	-	10	-	F29N	-	PCDDS	-	MDB	-	-	-	01	-	-
F10A2	-	20	-	F29R	-	PCDDS	-	RIO	-	-	-	10	-	-
F10A2	-	20	-	F28H	-	TMAH-O.063%	-	VMB	-	1	-	02	-	01
F10A2	-	20	-	F28H	-	TMAH-O.063%	-	VMB	-	1	-	02	-	02
F10A2	-	20	-	F28H	-	TMAH-O.063%	-	VMB	-	1	-	02	-	03
F10A2	-	20	-	F28H	-	TMAH-O.063%	-	VMB	-	1	-	02	-	04

A	Remarks
C10A2-10-F29N-H2O2-30%-SCME-01	Chemical Specific example with Room number
C10A2-10-F29N-IPA-CCB-01	Chemical Specific example with Room number
F10A2-20-F29T-NH4OH-29%-VMB-202	Chemical Specific example with gridline number
F10A2-20-F29N-IPA-VMB-106	Chemical Specific example with gridline number
F10A2-20-F43T-IPA-VMB-301	Chemical Specific example with gridline number
C10A2-10-F29N-PCDDS-EMOPNL-01	Common for overall Chemical system example with gridline number
C10A2-10-F29N-PCDDS-MDB-01	Common for overall Chemical system example with gridline number
F10A2-10-F29R-PCDDS-RIQ-10	Common for overall Chemical system example with gridline number
F10A2-20-F28H-TMAH-O.063%-VMB-102-01	Chemical specific Tagging inside VMB stick
F10A2-20-F28H-TMAH-O.063%-VMB-102-02	Chemical specific Tagging inside VMB stick
F10A2-20-F28H-TMAH-O.063%-VMB-102-03	Chemical specific Tagging inside VMB stick
F10A2-20-F28H-TMAH-O.063%-VMB-102-04	Chemical specific Tagging inside VMB stick

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6.5 System Abbreviation

SYSTEM ABBREV	SYSTEM IDENTIFIER
AA	ACETIC ACID
ACF	ACTIVATED CARBON FILTERS
ACS	ACCESS CONTROL SYSTEM
AC010	JSR CMS AC010
ACE	ACE
ACE002	ACE002
AD-10(TMAH)-2.38%)	TETRAMETHYL AMMONIUM HYDROXIDE WITH SURFACTANT
AE	ACID EXHAUST
AEAS	AREA EVACUATION ALARM SYSTEM
AHV	ARSENIC HOUSE VACUUM
AME	AMMONIA EXHAUST OR SCRUBBER
AMW	AMMONIA WASTE WATER
APM	AMMONIA PEROXIDE WASTEWATER
AWL	ACID WASTEWATER LOW
ANSC	ANTISCALANT
APC	AIR PARTICLE COUNTER
AR	ARGON (Ar)
ARR	AQUIFER RECHARGE AND RECOVERY
ASW	ARSENIC WASTE
AV	AUDIO VISUAL
AWR	ACID WASTE RECLAIM
BA	BREATHING AIR
BA	BREATHABLE AIR
BCL3	BORON TRICHLORIDE
BD	BENCH DRAIN
BF3	BORON FLUORIDE
BGW	BACKGRIND WASTE
BIOC	BIOCIDE
BLDG	BUILDING
C2H2	ACETYLENE
C3F8	OCTAFLUOROPROPANE
CACL2	CALCIUM CHLORIDE
CAOH	LIME SLURRY
CCE	CHAMBER CLEAN EXHAUST
CCTV	CLOSED CIRCUIT TELEVISION SYSTEM
CCW	CONCENTRATED COPPER WASTE
CD	CONDENSATE DRAIN

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SYSTEM ABBREV	SYSTEM IDENTIFIER
CDA	COMPRESSED DRY AIR
CHW	CHILLED WATER
CLPS	CLEAN LOW PRESSURE STEAM
CMD	CONCENTRATED METAL DRAIN
CMW	CONCENTRATED METALS WASTE
CO2	CARBON DIOXIDE (CO2)
CR	CONDENSATE RETURN
CSW	CORROSIVE SOLVENT WASTE
CuSO4-A	COPPER SULPHATE - ANOLITE (MCPQ-45L)
CuSO4-C	COPPER SULPHATE - CATHOLITE (MCPQ-71-02)
CW	CONDENSER WATER
DEV4262(TMAH-2.38%)	DILUTED TETRAMETHYL AMMONIUM HYDROXIDE WITH SURFACTANT
DILUTED AC010	JSR CMS AC010
DSP	MIXED ACID
DCW	DILUTE COPPER WASTE
DIO3	DEIONIZED OZONE WASTEWATER
DIR	DEIONIZED WATER RETURN
DIRC	DI RECLAIM
DIS	DEIONIZED WATER SUPPLY
DIW	DEIONIZED WATER
EAS	EMERGENCY ALARM SYSTEM
EDI	ELECTRODIONIZATION
EES	EMERGENCY EXHAUST SYSTEM
ENV	ENVIRONMENT MONITORING
EPOC	METALS WASTE EPOC MICROFILTRATION
ES	ELECTRICAL SYSTEM
EUVR-DEV	EUVR DEVELOPER
EUVR-DEV A04	EUVR DEVELOPER A04
EUVR-RINSE	EUVR RINSE 8
FAS	FIRE ALARM SYSTEM
FE	FUME EXHAUST
FECL3	FERRIC CHLORIDE
FMCS	FACILITY MONITORING CONTROL SYSTEM
FMW	FAB METALS WASTE
FN-DP001	FN-DP001 (BA)
FN-RP002	FN-RP002 (MIBC)
FPRO	FIRST PASS REVERSE OSMOSIS
FPSP	FIRE SPRINKLER SYSTEM
FPSU	FIRE SUPPRESSION SYSTEM
FPW	FIRE PROTECTION WATER
GCD	GOLD CONCENTRATE DRAIN

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SYSTEM ABBREV	SYSTEM IDENTIFIER
GCHW	GLYCOL CHILLED WATER
GE	GENERAL EXHAUST
GHVAC	GENERAL HEATING VENTILATING AND AIR CONDITIONING
GHG	GREENHOUSE GAS
GHW	GLYCOL HOT WATER
GTM	GAS TRANSFER MEMBRANE
H2	HYDROGEN (H2)
H2SO4	SULFURIC ACID
H3NSO3	SULFAMIC ACID
H2O2-30%	HYDROGEN PEROXIDE 30%
H2SO4-96%	SULFURIC ACID 96%
H3PO4-85%	PHOSPHORIC ACID 85%
HC-100	HC-100
HCL-36%	HYDROCHLORIC ACID 36%
HF100:1	HYDROFLUORIC ACID, DILUTED 100:1
HF120	HYDROFLUORIC ACID, DILUTED 20:1
HF-49%	HYDROFLUORIC ACID
HMDS	HEXAMETHYL DISILAZANE
HNO3-70%	NITRIC ACID 70%
HBW	HOST BIB WATER
HCL	HYDROCHLORIC ACID
HCV	HOUSE CLEANING VACUUM
HE	HELIUM (He)
HERO	HIGH EFFICIENCY REVERSE OSMOSIS
HEX	SODIUM HEXAMETAPHOSPHATE
HF	HYDROFLUORIC ACID
HFW	HYDROFLUORIC WASTEWATER
HMDW	HUMIDIFICATION WATER
HPCDA	HIGH PRESSURE COMPRESSEDLEAN DRY AIR
HPS	HIGH PRESSURE STEAM
HPW	HOT POTABLE WATER
HRR	HIGH RESISTIVITY WATER RETURN
HRS	HIGH RESISTIVITY WATER SUPPLY
HTRACE	HEAT TRACE
HUPW	HOT ULTRAPURE WATER
HV	HOUSE VACUUM
HVAC	HEATING VENTILATING AND AIR CONDITIONING
I-C6H8O7	(IND) CITRIC ACID
I-H2O2-30%	(IND) HYDROGEN PEROXIDE-30%
I-H2SO4-96%	(IND) SULFURIC ACID-96%
I-H3NSO3	(IND) SULFAMIC ACID

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SYSTEM ABBREV	SYSTEM IDENTIFIER
I-HCl-36%	(IND) HYDROCHLORIC ACID-36%
I-NaCl	(IND) SODIUM CHLORIDE
I-NaHSO ₃	(IND) SODIUM BISULFITE
I-NaOCl-12.5%	(IND) SODIUM HYPOCHLORITE-12.5%
I-NaOH-50%	(IND) SODIUM HYDROXIDE-50%
IA	INSTRUMENT AIR
ICA	INSTRUMENT COMPRESSED AIR
IHW	INDUSTRIAL HOT WATER
IPA	ISOPROPYL ALCOHOL (IPA)
IRR	IRRIGATION WATER
IW	INDUSTRIAL WASTEWATER
IWN	INDUSTRIAL WASTEWATER NEUTRALIZATION
LA6500	BUFFERED HYDROGEN FLUORIDE, LA6500
LAL30	BUFFERED HYDROGEN FLUORIDE, LAL30
LAL800	BUFFERED HYDROGEN FLUORIDE, LAL800
LCHe	LEAK CHECK HELIUM (LCHe)
LH ₂	LIQUID HYDROGEN (LH ₂)
LN ₂	LIQUID NITROGEN (LN ₂)
LO ₂	LIQUID OXYGEN (LO ₂)
LPA	LOW PRESSURE AIR
LPS	LOW PRESSURE STEAM
MUA	MAKE-UP AIR
MFW	METALS FINISHING WASTE
MIW	MONITORED INDUSTRIAL WASTE WATER
MO ₂ -0.2%	MO ₂ DILUTE (0.2%) CU SLURRY CLEANER
MO ₂ -5%	MO ₂ (5% CONCENTRATE)
MPS	MEDIUM PRESSURE STEAM
N ₂	NITROGEN (N ₂)
NaCl	SODIUM CHLORIDE SOLUTION
NaOCl	SODIUM HYPOCHLORITE
NaOH	SODIUM HYDROXIDE
NCD	NITRIC CLEAN DRAIN
NG	NATURAL GAS
NGV	NATURAL GAS VENT
NiZnCD	NICKEL/ZINC CONCENTRATE DRAIN
NH ₄ OH-2%	DILUTED 2% AMMONIUM HYDROXIDE
NH ₄ OH-29%	AMMONIUM HYDROXIDE 29%
NPR	NON-POTABLE RE-USE WATER
NPW	NON-POTABLE WATER
O ₂	OXYGEN (O ₂)
OFL	OVERFLOW LEADER

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SYSTEM ABBREV	SYSTEM IDENTIFIER
OPD25C(TMAH-25%)	TETRAMETHYL AMMONIUM HYDROXIDE 25%
OPD4000SC	SURFACTANT
OSW	OFF-SPEC WATER
PA	PUBLIC ADDRESS
PAC	POLYALUMINUM CHLORIDE
PCW	PROCESS COOLING WATER
POL	POLYMER
POLMB	POLISH MIXED BED
POLRO	POLISH REVERSE OSMOSIS
POU	POINT OF USE SYSTEM
PRO	PRIMARY REVERSE OSMOSIS PRODUCT
PAr	PROCESS ARGON
PH2	PROCESS HYDROGEN
PO2	PROCESS OXYGEN
PSW	PHOTO SOLVENT WASTE
PV	PROCESS VACUUM
PW	POTABLE WATER
RA	RETURN AIR
RAW	RAW WATER
RDL	ROOF DRAIN (RAINWATER) LEADER
RECLAIM	RECLAIM
RER600	PGMEA
RER900	CYCLOHEXANONE
RSC-T17	SURFER
ROCL	REVERSE OSMOSIS CLEANER
ROR	REVERSE OSMOSIS REJECT WATER
ROW	REVERSE OSMOSIS WATER
RRO	RECLAIM REVERSE OSMOSIS
RRRO	REJECT RECOVERY REVERSE OSMOSIS
RTWR	REMOTE TOOL WATER RETURN
RTWS	REMOTE TOOL WATER SUPPLY
RW	RIVER WATER
RWRO	RAW WATER REVERSE OSMOSIS
SA	SUPPLY AIR
SAW	SAW WASTE
SMBS	SODIUM METABISULFITE
SCHW	SECONDARY CHILLED WATER
SD	STORM DRAIN
SEPOC	SILICA WASTE EPOC MICROFILTRATION
SE	SOLVENT EXHAUST
SFS	SAND FILTER SYSTEM

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SYSTEM ABBREV	SYSTEM IDENTIFIER
SGHW	SNOW MELT GLYCOL HOT WATER
SiH4	SILANE (SiH4)
SIW	SILICA WASTEWATER
SLW	SLURRY WASTEWATER
SMW	SCRUBBER MAKEUP WATER
SOD	SOD WASTE
SOX	SOFTENED WATER
SPC-124	DILUTED SURFACTANT, SPC-124A
SST-A47	NMP MIXTURE, STRIPPER
SPRO	SECOND PASS REVERSE OSMOSIS
SS	SANITARY SEWER
SW	SOLVENT WASTE
TCA	TRANSFORMER CONTAINMENT AREA
TCHW	TERTIARY CHILLED WATER
TCW	TEST COOLING WATER
TEW	TEMPERED EYE WASH
TGMS	TOXIC GAS MONITORING SYSTEM
TMAH	TMAH
TMAH(0.063%)	TETRAMETHYL AMMONIUM HYDROXIDE DILUTE 0.063%
TPW	TEMPERED POTABLE WATER
TRW	TREATED RIVER WATER
UF	ULTRA FILTRATION
UN2	UTILITY NITROGEN (UN2)
UO2	UTILITY OXYGEN (UO2)
UPW	ULTRA PURE WATER
VOC	VOC
VSES	VENTILATION SYSTEM EMERGENCY SHUTDOWN
WAC	WEAK ACID CATION
WACPW	WAC PRODUCT WATER
WAP	WIRELESS ACCESS POINT
WAr	WELD ARGON (WAr)
WMFS	WATER MIST FIRE SUPPRESSION
WW	WASTE WATER
WWR	WASTE WATER RECLAIM
XCDA	EXTRA CLEAN DRY AIR
ZRP	ZERO REFERENCE PRESSURE
3DMAS	Hexamethylsilanetriamine
3MS	Trimethylsilane
APHP	AP-HP
Ar/Ne	Argon Mix
Ar/Xe/Ne	Argon 3.5% / Xenon 0.001% / Neon

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SYSTEM ABBREV	SYSTEM IDENTIFIER
B2H6 0.1% / H2	0.1% Diborane / 99.9% Hydrogen
B2H6 10% / H2	10% Diborane / 90% Hydrogen
B2H6 5% / N2	5% Diborane / 95% Nitrogen
C3H6	Propylene
C4F6	Hexafluoro-1,3-ButadieneHexafluoro-1,3-Butadiene
C4F8	Octafluorocyclobutane
CF4	Tetrafluoromethane
CH2F2	Difluoromethane
CH3F	Fluoromethane
CH4	Methane
CH4 10% / H2	10% Methane / 90% Hydrogen
CHF3	Trifluoromethane
Cl2	Chlorine
ClF3	Chlorine Trifluoride
CO	Carbon Monoxide
COS	Carbonyl Sulfide
F2 20% / N2	20% Fluorine / 80% Nitrogen
F2/Ar/Ne	Fluorine 1% / Argon 3.5% / Neon
F2/Kr/Ne	Fluorine 1% / Krypton 1.2% / Neon
GeH4 10% / H2	10% Germane / 90% Hydrogen
H2 2%/ Ar	Hydrogen 2% Argon Mix
H2 3%/N2	3% Hydrogen / Nitrogen Mix
H2 3.8% / N2	3.8% Hydrogen / 96.2% Nitrogen
HBr	Hydrogen Bromide
HCDS	Hexachlorodisilane
HCl	Hydrogen Chloride
He 1.2% / N2	1.2% Helium / 98.8% Nitrogen
HF (5N)	Hydrogen Fluoride (5N)
HF (clean)	Hydrogen Fluoride (clean)
HPCO2	High Pressure Carbon Dioxide
HPH2	High Pressure Hydrogen
HPN2	High Pressure Nitrogen
Kr	Krypton
Kr 1.25% / Ne	Krypton 1.25% / 98.75% Neon
LTO520	LTO520
N2O	Nitrous Oxide
NF3	Nitrogen Trifluoride
NH3 (Ind)	Ammonia (Ind)
NH3 (Process)	Ammonia (Process)
NZERO	N-ZERO (Organosilane)
OMCTS	Octamethylcyclotetrasiloxane

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SYSTEM ABBREV	SYSTEM IDENTIFIER
Owyhee	Owyhee
PH3	Phosphine
PH3 1%/He	1% Phosphine / Helium Mix
PH3 5%/N2	5% Phosphine / Nitrogen Mix
PHe	Process Helium
Sawtooth	Sawtooth
SF6	Sulfur Hexafluoride
Si2H6	Disilane
SiCl4	Silicon Tetrachloride
SiH2Cl2	Dichlorosilane
SiH4	Silane
SO2	Sulfur Dioxide
TEB	Triethyl Borate
TEOS	Tetraethoxysilane
TiCl4	Titanium (IV) Tetrachloride
TMA	Trimethylaluminum
TMS	Tetramethylsilane
WF6	Tungsten Hexafluoride
Xe	Xenon
Yamuna	YAMUNA (Organosilane)
ZnSO4	ZINC SULFATE SOLUTION

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- 6.6 Equipment Abbreviation. User choices are allowed to use, however the these should be discussed during design.

EQUIP ABBREV	EQUIPMENT
ACF	ACTIVATED CARBON FILTER
ACFS	AIRGARD CYCLONE FUME SCRUBBER
ACT	ACTUATOR
ACU	AIR CONDITIONER UNIT
AD	AIR DRYER
ADR	AUTOMATIC DOOR
AHU	AIR HANDLING UNIT
AIRCOMP	AIR COMPRESSOR
AIRREC	AIR RECEIVER
AIRSHR	AIR SHOWER
AS	AUTOSHUTDOWN RACK/CABINET
ATS	AUTOMATIC TRANSFER SWITCH
BAS	BREATHABLE AIR STATION
BCTU	BULK CHEMICAL TRANSFER UNIT
BD	BUILDING ISOLATION VALVE
BDTNK	BLEND & DISTIRBUTION TANK
BFD	BACK FLOW DEVICE
BKR	BREAKER
BLO	BLOWER
BLR	BOILER
BPMP	BOOSTER PUMP
BPR	BACK PRESSURE REGULATOR
BTNK	BULK TANK
BUSS	BUSSWAY
BUSSDUCT	BUSS DUCT
CATNK	COMPRESSED AIR TANK
CBU	CHEMICAL BLEND UNIT
CCB	CLEAN COUPLING BOOTH
CDM	CHEMICAL DISPENSE MODULE
CC	COOLING COIL
CCEF	CHAMBER CLEAN EXHAUST FAN
CENT	CENTRIFUGE
CH	CHILLER
CHAR	CHARGER
CKV	CHECK VALVE
CLNTR	CHLORINATOR
CLPSG	CLEAN LOW PRESSURE STEAM GENERATOR
CLR	COOLER

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EQUIP ABBREV	EQUIPMENT
CMPTR	COMPUTER
CNDU	CONDENSOR UNIT
COIL	COIL
COMP	COMPRESSOR
CONDPMP	CONDENSATE PUMP
CONDREC	CONDENSATE RECEIVER
CRAH	COMPUTER ROOM AIR HANDLER
CRAC	COMPUTER ROOM AIR CONDITIONER
CRMAU	CLEANROOM MAU
CT	COOLING TOWER
CTFLTR	COOLING TOWER FILTRATION SYSTEM
CTDM	CHEMICAL TRANSFER & DISPENSE MODULE
CTNK	CONCENTRATE TANK
CTU	CHEMICAL TRANSFER UNIT
CuSO4-A	COPPER SULPHATE - ANOLITE (MCPQ-45L)
CuSO4-C	COPPER SULPHATE - CATHOLITE (MCPQ-71-02)
CV	CONTROL VALVE
CY	CRYOGENIC CYLINDER
DA	DEAERATOR
DC	DUCT COIL
DCC	DRY COOLING COIL
DDC	DIRECT DIGITAL CONTROLLER
DDCPNL	DIRECT DIGITAL CONTROL PANEL
DE	DEWAR
DEG	DEGASIFIER
DEIONIZER	DEIONIZER
DFOUNT	DRINKING FOUNTAIN
DHC	DEHUMIDIFIER COIL
DIFF	DIFFUSER
DLV	DOCK LEVELER
DMHUM	DUCT-MOUNTED HUMIDIFIER
DMPR	DAMPER
DR	DOOR
DRCLR	DRY COOLER
DRN	DRAIN
DRYER	DRYER
DS	DOWN SPOUT
DTNK	DAY TANK
DUCTFUR	DUCT FURNACE
DUPS	DIESEL UPS
DVACPMP	DEGASSER VACUUM PUMP

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EQUIP ABBREV	EQUIPMENT
DW	DUMBWAITER
ECU	EVAPORATIVE COOLING UNIT
EDI	ELECTRODEIONIZATION
EEF	EMERGENCY EXHAUST FAN
EHTR	ELECTRIC HEATER
ELECPNL	ELECTRICAL PANEL
ELEV	ELEVATOR
EL	EMERGENCY LIGHTING
ENG	ENGINE
EPS	EMERGENCY POWER SYSTEM
EVAP	EVAPORATOR
EVB	EQUIPMENT VALVE BOX
EWIN	ELECTROWINNER
EWS	EYE WASH STATION
EWSD	EARLY WARNING SMOKE DETECTION
EXF	EXHAUST FAN
EXV	EXPANSION VALVE
FAN	FAN
FAPNL	FIRE ALARM PANEL
FATC	FIRE ALARM TERMINATION CABINET
FBA	FLASHBACK ARRESTOR
FCB	FAN COIL BLOCK
FCO	FLOOR CLEAN OUT
FCU	FAN COIL UNIT
FCUDX	FAN COIL UNIT W/ DX COOLING
FDR	FEEDER
FFU	FAN FILTER UNIT
FHV	FIRE HOSE VALVE
FLTR	FILTER
FLTRPRS	FILTER PRESS
FLTRVSL	FILTER VESSEL
FM	FLOW MONITOR/METER
FMCS	FACILITIES MONITORING CONTROL SYSTEM
FRUDR	FIRE ROLL UP DOOR
FTK	FLASH TANK
GA	GAS ANALYZER
GASBOX	GAS BOX W/ EQUIP FOR MULTIPLE DIST. SYSTEMS
GB	GAS BLENDER
GC	GAS CABINET
GCHW	GLYCOL CHILLED WATER
GDB	GAS DISTRIBUTION BOX/PANEL

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EQUIP ABBREV	EQUIPMENT
GEN	GENERATOR
GEXF	GENERAL EXHAUST FAN
GF	GAS FILTER
GHWPMP	GLYCOL HOT WATER PUMP
GI	GAS ISOLATION BOX/PANEL
GP	GAS PANEL
GRDN	GUARDIAN
GS	GAS STICK
GTM	GAS TRANSFER MEMBRANE
HEL	HELIOS
HOV	HAND OPERATED VALVE
HRPMP	HEAT RECLAIM PUMP
HRU	HEAT RECOVERY UNIT
HRW	HEAT RECOVERY WATER
HST	HOIST
HTC	HEATING COIL
HTEXF	HEAT EXHAUST FAN
HTR	HEATER
HTRACE	HEAT TRACE
HUM	HUMIDIFIER
HVPMP	HOUSE VACUUM PUMP
HWBLR	HOT WATER BOILER
HX	HEAT EXCHANGER
HYDRANT	HYDRANT (FIRE)
ICEM	ICE MACHINE
INJ	INJECTOR
IWIN	IONWINNER
JB	JUNCTION BOX
JTB	JOINT BOX
IX	ION EXCHANGE VESSEL
KRO	KRONIS SCRUBBER
LA	LIQUID ANALYZER
LB	LIQUID BLENDER
LC	LIQUID CABINET
LD	LEAK DETECTION
LF	LIQUID FILTER
LFLTR	LINT FILTER
LGT	LIGHT
LIFT	LIFT
LO	LIQUID ISOLATION BOX/PANEL
LP	LIQUID PANEL

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EQUIP ABBREV	EQUIPMENT
LS	LIQUID STICK
LSR	LOCAL SCRUBBER RECLAIM
LT	LATERAL ISOLATION VALVE
LTCHW	LOW TEMPERATURE CHILLED WATER
LX	LIQUID DISTRIBUTION BOX/PANEL
MAU	MAKE-UP AIR UNIT
MB	MIXED BED DEIONIZATION VESSEL
MCC	MOTOR CONTROL CENTER
MF	MICRO FILTER
MOD	MOTOR OPERATED DAMPER
MPSCRB	MULTIPURPOSE SCRUBBER
MTCHW	MEDIUM TEMPERATURE CHILLED WATER
MTR	MOTOR
MXR	MIXER
NGM	NATURAL GAS METER
NL	NORMAL LIGHTING
NPS	NORMAL POWER SYSTEM
NTNK	NEUTRALIZATION TANK
OA	OUTSIDE AIR
OPR	OPERATOR
ORFC	ORIFICE
OSS	OIL/SEDIMENT SEPARATOR
PC	PARTICLE COUNTER
PCDMPR	PRESSURE CONTROL DAMPER
PDP	POWER DISTRIBUTION PANEL
PHC	PREHEAT COIL
PHPMP	PREHEAT PUMP
PLC	PROGRAMMABLE LOGIC CONTROLLER
PLCPNL	PROGRAMMABLE LOGIC CONTROLLER PANEL
PMB	PRIMARY MIXED BED VESSEL
PMP	PUMP
PNL	PANEL
PNLVPMP	PANEL VACUUM PUMP
POL	POLISHER
POS	POSITION
PS	POWER SUPPLY
PSA	PRESSURE SWING ABSORPTION
PU	PURIFIER
PUPS	PORTABLE UPS
PVPMP	PROCESS VACUUM PUMP
RA	RETURN AIR

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EQUIP ABBREV	EQUIPMENT
RAF	RETURN AIR FAN
RC	REACTOR-CLARIFIER
RCVR	RECEIVER
REFRIG	REFRIGERATOR
REG	REGULATOR
RF	RECIRCULATION FAN
RHPMP	REHEAT PUMP
RIO	REMOTE I/O
RK	RACK
RNPMP	RUNAROUND PUMP
RO	REVERSE OSMOSIS FILTER
ROPMP	REVERSE OSMOSIS HIGH PRESSURE PUMP
RPU	REFRIGERANT PURGE UNIT
RTU	ROOF TOP UNIT
RUDR	ROLL UP DOOR
SA	SUPPLY AIR
SAC	STRONG ACID CATION VESSEL
SAFSHR	SAFETY SHOWER
SCRB	SCRUBBER
SEPR	SEPARATOR
SEXF	SOLVENT EXHAUST FAN
SF	SEWER FLUME
SHCLNR	SHOE CLEANER
SLPMP	SLUDGE PUMP
SLTNK	SLUDGE TANK
SMB	SECONDARY MIXED BED
SMS	SAG MITIGATION SYSTEM
SPS	STANDBY POWER SYSTEM
SUPS	STATIC UPS
SOX	SOFTENER
SP	SOURCE PLATE
SPRS	STEAM PRESSURE REDUCING STATION
SRF	STATIC RELIEF FAN
SSFVSL	SIDE STREAM FILTER VESSEL
ST	SITE ISOLATION VALVE
STNK	SEDIMENTATION TANK
STU	STORAGE UNIT
SUMP	SUMP
SPS	SUMP PUMP
SW	SWITCHING BOX/PANEL
SWBD	SWITCH BOARD

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EQUIP ABBREV	EQUIPMENT
SWPMP	SEAL WATER PUMP
TB	TERMINAL BOX
TC	TWIN COLUMN DEIONIZATION VESSEL
TCS	THERMAL CONDITIONING SYSTEM
TD	TRENCH DRAIN
TELE	TELECOMUNICATION
TL	TOOL ISOLATION VALVE
TMP	TEMPEST
TNK	TANK
TPU	THERMAL PROCESSING UNIT
TRCMP	TRASH COMPACTOR
TUR	TURBINE
UF	ULTRA FILTER
UHTR	UNIT HEATER
UPS	UNINTERRUPTIBLE POWER SUPPLY UNIT
UPSBAT	UPS BATTERY
UV	ULTRAVIOLET STERILIZER/TOC REDUCTION
V	VALVE
VAC	VACUUM
VACPMP	VACUUM PUMP
VACSEP	VACUUM SEPARATOR
VAP	VAPORIZER
VAV	VARIABLE AIR VOLUME BOX
VFD	VARIABLE FREQUENCY SPEED DRIVE
VMB	VALVE MANIFOLD BOX
VOC	VOLATILE ORGANIC CHEMICALS ABATEMENT UNIT
VOCPF	VOC PROCESS FAN
WAC	WEAK ACID CATION
WFLTR	WATER FILTER
WHTR	WATER HEATER
WSOFT	WATER SOFTENER
WTS	WATER TRAP SILENCER
TR	TRANSFORMER

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- 6.7 Instrument Abbreviation. User choices are allowed to use, however the these should be discussed during design.

INSTRUMENT ABBREV	INSTRUMENT
ACV	ANALYSIS CONTROL VALVE
AE	ANALYSIS SENSING ELEMENT
AI	ANALYSIS INDICATOR
AIC	ANALYSIS INDICATING CONTROLLER
AIT	ANALYSIS INDICATING TRANSMITTER (PH, CONDUCTIVITY, ORP, RESISTIVITY)
AY	ANALYSIS CONTROL VALVE I/P OR SOLENOID
CE	CONDUCTIVITY SENSING ELEMENT
CI	CONDUCTIVITY INDICATOR
CIC	CONDUCTIVITY INDICATING CONTROLLER
DSP	DUCT STATIC PRESSURE
E/H	ELECTRO / HYDRAULIC CONVERTER
E/I	VOLTAGE / CURRENT CONVERTER
FCV	FLOW CONTROL VALVE
FE	FLOW ELEMENT
FG	FLOW GLASS
FI	FLOW INDICATOR
FIC	FLOW INDICATING CONTROLLER
FIT	FLOW INDICATING TRANSMITTER
FQ	FLOW TOTALIZER
FQIT	FLOW TOTALIZING INDICATION TRANSMITTER
FQY	FLOW TOTALIZING COMPUTER
FS	FLOW SWITCH
FSH	FLOW SWITCH HIGH
FSL	FLOW SWITCH LOW
FT	FLOW TRANSMITTER
HCV	HUMIDITY CONTROL VALVE
HE	HUMIDITY SENSING ELEMENT
HI	HUMIDITY INDICATOR
HIC	HUMIDITY INDICATING CONTROLLER
HOA	HAND / OFF / AUTO SWITCH
HS	HAND SWITCH
HS01A	"STARTER" HAND
HS01B	"STARTER" AUTO
HT	HUMIDITY TRANSMITTER
I/E	CURRENT TO VOLTAGE CONVERTER
I/I	CURRENT TO CURRENT CONVERTER
I/P	CURRENT TO PNEUMATIC CONVERTER

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INSTRUMENT ABBREV	INSTRUMENT
IE	CURRENT SENSING ELEMENT
IS	CURRENT SWITCH
JSL	POWER SWITCH LOW
KV	TIMER VALVE
KY	TIMER VALVE SOLENOID
LCV	LEVEL CONTROL VALVE
LE	LEVEL SENSING ELEMENT
LG	LEVEL GLASS
LI	LEVEL INDICATOR
LIC	LEVEL INDICATING CONTROLLER
LIT	LEVEL INDICATING TRANSMITTER
LPC	LIQUID PARTICLE COUNTER
LS	LEVEL SWITCH
LSH	LEVEL SWITCH HIGH
LSHH	LEVEL SWITCH HIGH HIGH
LSL	LEVEL SWITCH LOW
LSLL	LEVEL SWITCH LOW LOW
LT	LEVEL TRANSMITTER
ME	MOISTURE SENSING ELEMENT
MI	MOISTURE INDICATOR
MIC	MOISTURE INDICATING CONTROLLER
MIT	HUMIDITY/MOISTURE INDICATING TRANSMITTER
MIT	OUTSIDE AIR HUMIDITY
MOV	MOTOR OPERATED VALVE
MS	MOISTURE / RH SWITCH
MT	MOISTURE / RH TRANSMITTER
MV	MOISTURE / RH VALVE
MY	MOISTURE / RH CONTROL
PCV	PRESSURE CONTROL VALVE
PDI	PRESSURE DIFFERENTIAL INDICATOR
PDIS	PRESSURE DIFFERENTIAL INDICATING SWITCH
PDIT	PRESSURE DIFFERENTIAL INDICATING TRANSMITTER
PDS	PRESSURE DIFFERENTIAL SWITCH
PDSH	PRESSURE DIFFERENTIAL SWITCH HIGH
PDT	PRESSURE DIFFERENTIAL TRANSMITTER/ROOM STATIC PRESSURE
PE	PRESSURE SENSING ELEMENT
PI	PRESSURE INDICATOR
PIC	PRESSURE INDICATING CONTROLLER
PIS	PRESSURE INDICATING SWITCH
PIT	PRESSURE INDICATING TRANSMITTER
PS	PRESSURE SWITCH

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INSTRUMENT ABBREV	INSTRUMENT
PSE	RUPTURE DISK
PSH	PRESSURE SWITCH HIGH
PSL	PRESSURE SWITCH LOW
PSV	PRESSURE SAFETY VALVE
PT	PRESSURE TRANSMITTER
QE	QUANTITY SENSING ELEMENT
QI	QUANTITY INDICATOR
QIT	QUANTITY INDICATING TRANSMITTER
QT	QUANTITY TRANSMITTER
QT	QUANTITY TRANSMITTER
RO	RESTRICTION ORIFICE
RSP	ROOM STATIC PRESSURE
RV	RELIEF VALVE
SC	SPEED CONTROL
SD	SMOKE DETECTOR
SE	SPEED SENSING ELEMENT
SI	SPEED INDICATOR
SIC	SPEED INDICATING CONTROLLER
SIC01A	VFD SPEED FEEDBACK
SIC01B	VFD SPEED COMMAND
SOV	SOLENOID VALVE
ST	SPEED TRANSMITTER
SV	SAFETY VALVE
TCD	TEMPERATURE CONTROL DAMPER
TCV	TEMPERATURE CONTROL VALVE
TE	TEMPERATURE SENSOR
TI	TEMPERATURE INDICATOR
TIC	TEMPERATURE IND. CONTROLLER
TIT	OUTSIDE AIR TEMPERATURE
TIT	TEMPERATURE IND. TRANSMITTER
TS	TEMPERATURE SWITCH
TSH	TEMPERATURE SWITCH HIGH
TSL	TEMPERATURE SWITCH LOW
TT	TEMPERATURE TRANSMITTER
TW	THERMOWELL
UR	UNIVERSAL RECORDER
VC	VIBRATION CONTROLLER
VE	VIBRATION SENSING ELEMENT
VR	VIBRATION RECORDER
VS	VIBRATION SWITCH
VT	VIBRATION TRANSMITTER

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INSTRUMENT ABBREV	INSTRUMENT
YA	STATUS ALARM
YA	VFD FAULT STATUS
YC	STATE CONTROL
YC01	VFD FAN START/STOP
YC01	"STARTER" START/STOP
YI	STATUS INDICATOR
YS	STATUS SWITCH, RELAY CONTACT
YS	VFD REMOTE STATUS
YS	VFD RUN STATUS
ZC	POSITION CONTROLLER
ZDR	POSITION DIFFERENTIAL RECORDER
ZE	POSITION SENSOR
ZR	POSITION RECORDER
ZS	POSITION SWITCH
ZSC	POSITION/LIMIT SWITCH CLOSED
ZSO	POSITION/LIMIT SWITCH OPEN
ZT	POSITION TRANSMITTER
ZUM	CONTROL DRIVE - MOTOR DRIVEN
ZUP	CONTROL DRIVE - PNEUMATIC

7.0 Document Control

7.1 **Approval:** GLOBAL_FAC_GFTT_APPR

7.2 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY

7.3 Final Viewer Access

All Micron Team Members at all Sites

7.4 Retention

Adherence to the requirements specified in the
[**Global Facilities - Document Control Procedures**](#)

7.5 Review

Biennially (two years)

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8.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	03/23/22
1	3.0	Acronym "LPOC" added	03/26/24
	6.1	Updated Location grid description, equipment numbering, and optional removal	
	6.2	Updated Location grid description, equipment numbering, and optional removal	
	6.4	Updated LPOC tagging	
	6.5, 6.6	Additional abbreviation	

Appendix A: Deviation List

Any deviation from this specification shall be submitted for approval during vendor evaluation. Deviation list format shall be the following:

Site: _____

No	Specification Section and Item	Deviation Proposal	Scope Impact (Benefit, Cost, Schedule)	Remark	Approval: 1. Site 2. GFTT